Employee Management System (EMS)

CS3009 – Software Engineering

---

1. Introduction

The Employee Management System (EMS) is a Java-based business management system designed to automate and centralize employee-related operations within an organization. Traditional manual record keeping and spreadsheet-based systems often result in data inconsistency, security vulnerabilities, payroll errors, and inefficient workforce tracking.

The EMS provides a secure, role-based platform that manages employee data, attendance, leave, payroll, documents, and reporting in a structured and scalable way.

Sprint 1 focuses on core authentication, employee management, leave handling, attendance, payroll, and reporting features.

The system is developed using the Scrum framework to ensure iterative delivery, stakeholder feedback, and continuous improvement.

---

2. Project Vision

The vision of EMS is to:

- Provide a secure and centralized employee management platform
- Automate HR processes
- Improve payroll accuracy
- Enhance transparency in leave and attendance
- Enable management-level decision-making using real-time dashboards
- Maintain digital documentation for compliance and record-keeping

The system aims to reduce manual errors, improve efficiency, and ensure secure access to sensitive employee information.

---

3. Intended Use of the System

3.1 Stakeholders

1. Administrator
2. HR Manager
3. Manager
4. Employee
5. Finance Department

6. Top Management
7. IT Support Team

3.2 Stakeholder Needs

Administrator

- Define roles and permissions
- Maintain system security
- Manage user accounts

HR Manager

- Manage employee records
- Handle payroll
- Manage documents
- Oversee attendance & leave

Manager

- Approve/reject leave
- View team dashboard
- Monitor attendance

Employee

- Login securely
- Apply for leave
- Mark attendance
- View profile and payroll

Finance Department

- Verify payroll reports
- Ensure salary accuracy

Top Management

- Access performance dashboards
- Monitor workforce trends

---

4. Overall System Features

The system will provide:

1. User Authentication
2. Role-Based Access Control (RBAC)
3. Employee CRUD Management
4. Employee Profile Viewing
5. Leave Management
6. Attendance Management
7. Payroll Generation
8. Employee Search
9. Document Upload & Storage
10. Dashboard & Reports

---

5. User Stories

US1 – User Login

As a registered user
I want to log into the system using email and password
So that I can access features according to my role

US2 – Role-Based Access Control

As an administrator
I want to restrict features based on user roles
So that unauthorized access is prevented

US3 – Add New Employee

As an HR manager
I want to register a new employee
So that records are stored

US4 – View Employee Profile

As an employee or HR manager
I want to view employee information
So that I can see job-related details

US5 – Update Employee Information

As an HR manager
I want to update employee details
So that records remain accurate

US6 – Deactivate Employee

As an HR manager
I want to deactivate an employee
So that login is disabled but history remains

US7 – Apply for Leave

As an employee
I want to submit leave requests
So that I can request time off

US8 – Approve/Reject Leave

As a manager
I want to approve or reject leave
So that availability is managed

US9 – Mark Daily Attendance

As an employee
I want to mark check-in/check-out
So that working hours are recorded

US10 – Generate Payroll

As an HR manager
I want to calculate salaries
So that payroll is accurate

US11 – Search Employee

As an HR manager
I want to search employees
So that I can quickly locate records

US12 – Upload Employee Document

As an HR manager
I want to upload documents
So that records are maintained digitally

US13 – View Dashboard / Reports

As a manager
I want to view key metrics
So that I can monitor workforce performance

---

6. Structured Specifications

6.1 User Authentication

Inputs: Email, Password
Processing:

- Validate inputs
- Retrieve user record
- Compare encrypted password
- Check account status
- Create session

Outputs:

- Dashboard redirect (Success)
- Error message (Failure)

6.2 Role-Based Access Control

Inputs: User Role
Processing:

- Identify role
- Load permitted modules
- Restrict unauthorized access

Outputs:

- Dynamic dashboard display

6.3 Add New Employee

Inputs: Name, Email, Phone, Department, Role, Salary
Processing:

- Validate required fields
- Generate Employee ID
- Store in database
- Create login credentials

Outputs:

- Success confirmation

6.4 Leave Application

Inputs: Employee ID, Leave Type, Start Date, End Date
Processing:

- Validate leave balance

- Store leave request
- Notify manager

Outputs:

- Leave submitted confirmation

6.5 Attendance Management

Inputs: Employee ID, Timestamp
Processing:

- Record check-in/check-out
- Calculate working hours

Outputs:

- Attendance confirmation

6.6 Payroll Generation

Inputs: Salary, Attendance, Overtime, Deductions
Processing:

- Calculate gross salary
- Apply bonuses
- Subtract deductions
- Generate payroll record

Outputs:

- Salary slip
- Payroll report

6.7 Document Upload

Inputs: Employee ID, File
Processing:

- Validate file format
- Store securely
- Link to employee profile

Outputs:

- Upload confirmation

---

7. Non-Functional Requirements (NFR)

Security

- Passwords shall be encrypted
- Role-based access control shall be enforced
- Session timeout after inactivity

Performance

- System shall respond within 3 seconds
- System shall support 500 concurrent users

Reliability

- System shall maintain 99% uptime
- Automatic database backup

Usability

- User-friendly interface
- Minimal training required

Maintainability

- Modular architecture
- Clean code documentation

Scalability

- System shall support future expansion

---

8. Scrum Board (Sprint 1)

Sprint 1 Modules:

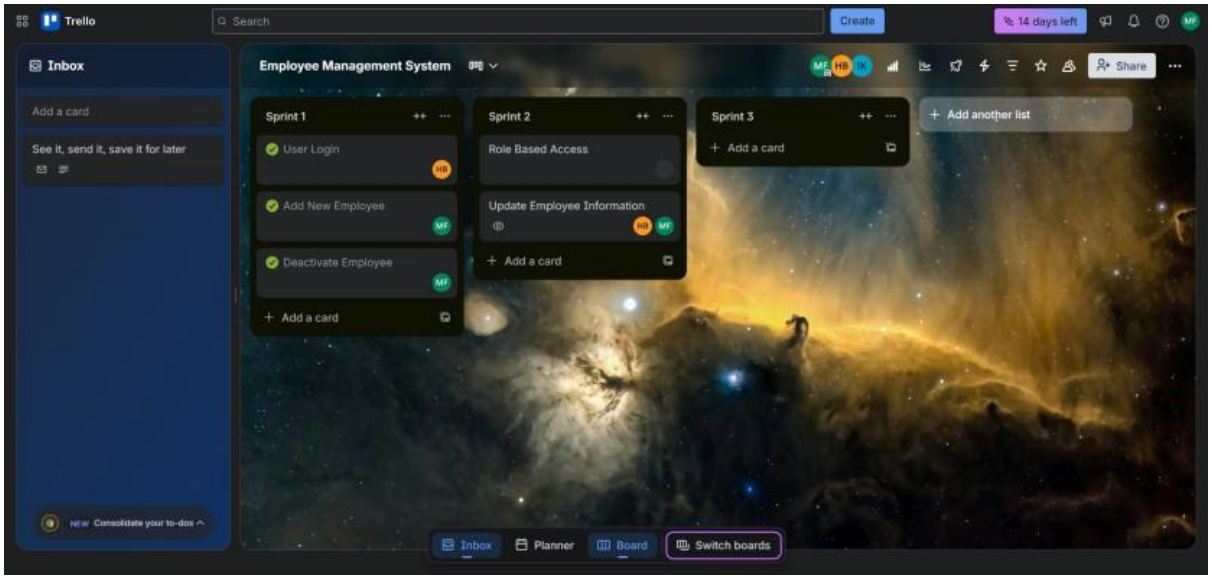
- User Login
- Add New Employee
- Deactivate Employee
- Role Based Access
- Update Employee Information

Tasks:

- Database design
- Backend implementation

- Frontend UI
- Testing
- Integration

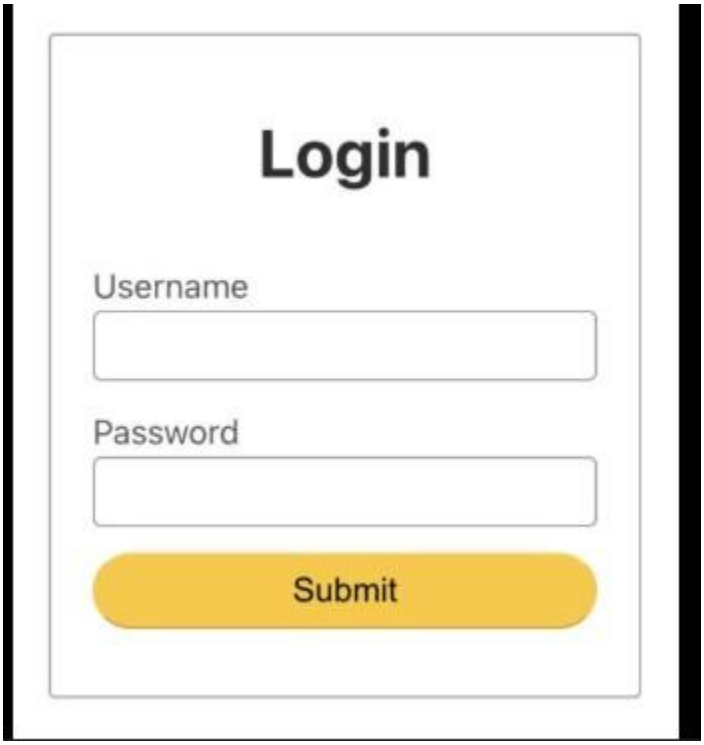
(Trello snapshots and burn-down chart is attached.)



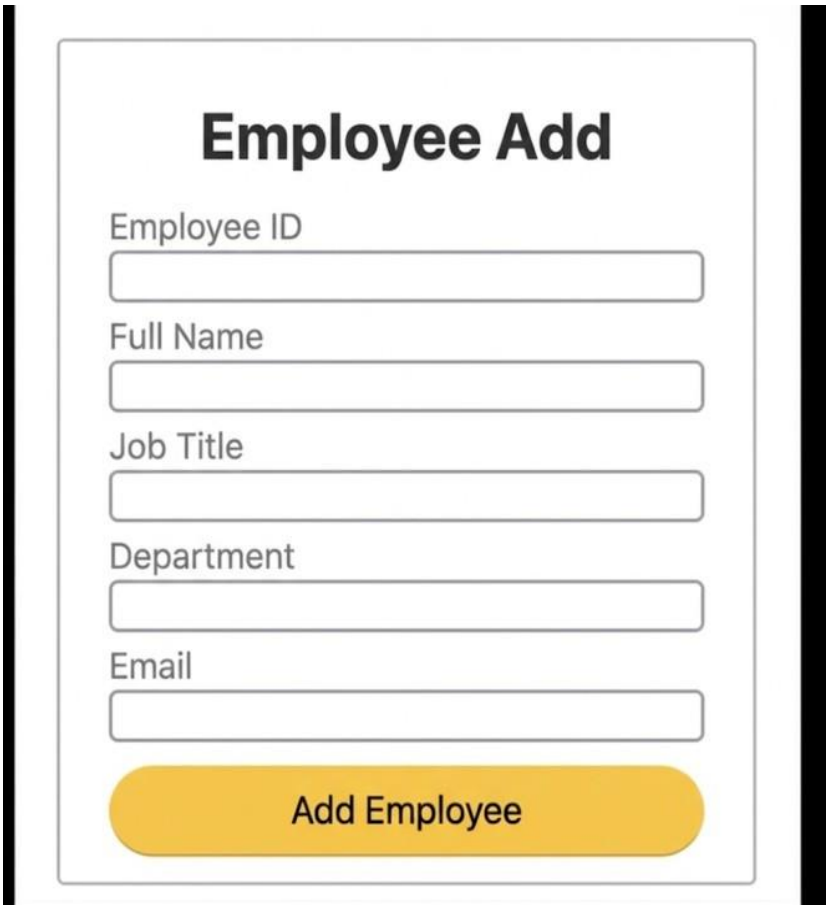
---

9. Iteration 1 Implementation Screenshots

- Login Page



- Add Employee Screen



- Employee Deactivate

---

10. Work Division

Muhammad Faheem:

- Scrum Master and Product Owner

Irtaza Kazmi:

- Development

Humayun Bilal:

- UI design

---

11. Conclusion

The Employee Management System provides a complete, secure, and scalable solution for modern organizations. Through Scrum-based iterative development, the system ensures continuous improvement and stakeholder alignment.